

Bayesian Probability*(find and work with a partner)*

Clinical Trials Example. Say the probability of a disease (D) in the general population is 1 in 100, i.e. $P(D) = \frac{1}{100}$. This is our *prior* probability of the disease (i.e., without any data).

Furthermore, say we have a test for this disease with 90% accuracy. We will call the results of the test positive (“pos”) and negative (“neg”). 90% accuracy means that $P(\text{pos} \mid D) = \frac{9}{10}$, and $P(\text{neg} \mid H) = \frac{9}{10}$, where H means healthy.

What we actually want to know is: what is the probability of having the disease, given a positive test?

1. Apply Bayes’ rule to $P(D \mid \text{pos})$. Recall that Bayes’ rule says:

$$P(A \mid B) = \frac{P(A)P(B \mid A)}{P(B)}.$$

2. We can often write the denominator as the sum of $P(a, B)$ for all options $a \in \text{vals}(A)$:

$$P(A \mid B) = \frac{P(A)P(B \mid A)}{\sum_{a \in \text{vals}(A)} P(a, B)}.$$

Use this idea to expand the denominator in the clinical trials example and compute a numerical value for $P(D \mid \text{pos})$.

Naive Bayes*(find and work with a partner)*

Say we have two tests for a specific disease. Each test (features f_1, f_2) can come back either positive “pos” or negative “neg”, and the true underlying condition of the patient is represented by y ($y = 1$ is “healthy” and $y = 2$ is “disease”). We observe this training data where $n = 7$ and $p = 2$:

$\mathbf{x}$	f_1	f_2	y
$\mathbf{x}_1$	pos	neg	1
$\mathbf{x}_2$	pos	pos	2
$\mathbf{x}_3$	pos	neg	2
$\mathbf{x}_4$	neg	neg	1
$\mathbf{x}_5$	pos	neg	2
$\mathbf{x}_6$	neg	neg	1
$\mathbf{x}_7$	neg	pos	2

1. To estimate the probability $p(y = k)$, for $k = 1, 2, \dots, K$, we will use the formula:

$$\theta_k = \frac{N_k + 1}{n + K},$$

where N_k is the count (“Number”) of data points where $y = k$. Compute θ_1 and θ_2 . What would θ_1 and θ_2 be if we in fact had *no* training data?

2. To estimate the probabilities $p(x_j = v \mid y = k)$ for all features j , values v , and class labels k , we will use the formula:

$$\theta_{k,j,v} = \frac{N_{k,j,v} + 1}{N_k + |f_j|},$$

where $N_{k,j,v}$ is the count of data points where $y = k$ and $x_j = v$, and $|f_j|$ is the number of possible values that f_j (feature j) can take on. Fill in the following tables with these θ values.

$y = 1$	pos	neg
f_1		
f_2		

$y = 2$	pos	neg
f_1		
f_2		

3. Continuing the example above, say we have a new data point $\mathbf{x}_{\text{test}} = [\text{neg}, \text{pos}]$. Our goal is to predict the class label based on the Naive Bayes posterior probability. In practice, we will compute

this probability for each class k , based on our estimates (θ_k and $\theta_{k,j,v}$ terms). Then we will assign this data point the class label with maximum probability:

$$\hat{y} = \arg \max_{k \in \{1,2,\dots,K\}} p(y = k \mid \mathbf{x}) = \arg \max_{k \in \{1,2,\dots,K\}} p(y = k) \prod_{j=1}^p p(x_j \mid y = k).$$

For this $\mathbf{x}_{\text{test}}$, compute $p(y = 1 \mid \mathbf{x}_{\text{test}})$ and $p(y = 2 \mid \mathbf{x}_{\text{test}})$ and then assign a prediction label $\hat{y}$.

4. For the tennis example below, fill in the $\theta_{k,j,v}$ terms (thinking about how this could be implemented using dictionaries).

Day	Outlook	Temperature	Humidity	Wind	PlayTennis (y)
$\mathbf{x}_1$	Sunny	Hot	High	Weak	No
$\mathbf{x}_2$	Sunny	Hot	High	Strong	No
$\mathbf{x}_3$	Overcast	Hot	High	Weak	Yes
$\mathbf{x}_4$	Rain	Mild	High	Weak	Yes
$\mathbf{x}_5$	Rain	Cool	Normal	Weak	Yes
$\mathbf{x}_6$	Rain	Cool	Normal	Strong	No
$\mathbf{x}_7$	Overcast	Cool	Normal	Strong	Yes
$\mathbf{x}_8$	Sunny	Mild	High	Weak	No
$\mathbf{x}_9$	Sunny	Cool	Normal	Weak	Yes
$\mathbf{x}_{10}$	Rain	Mild	Normal	Weak	Yes
$\mathbf{x}_{11}$	Sunny	Mild	Normal	Strong	Yes
$\mathbf{x}_{12}$	Overcast	Mild	High	Strong	Yes
$\mathbf{x}_{13}$	Overcast	Hot	Normal	Weak	Yes
$\mathbf{x}_{14}$	Rain	Mild	High	Strong	No

y=No (0)

outlook	Sunny:	Overcast:	Rain:
temperature	Cool:	Mild:	Hot:
humidity	Normal:	High:	
wind	Weak:	Strong:	

y=Yes (1)

outlook	Sunny:	Overcast:	Rain:
temperature	Cool:	Mild:	Hot:
humidity	Normal:	High:	
wind	Weak:	Strong:	